

Element C

After researching the problem as seen in element A and B along with the comments from Mrs. Moody, we decided on some basic guidelines for our design. Our criteria and constraints are listed below and are in order of importance with the first being the most important and decreasing in importance as you move down.

Criteria:

1. Petals Fall:
 - a. Must have at least 3 instances where one or more petals fall at specific times set/initiated by the tech crew of the musical.
2. Lights:
 - a. Must have at least 3 individual lights aimed at the rose from below the bulb.
3. Color:
 - a. Must have red petals and a green stem.
4. Hight:
 - a. Must be at least 1.5ft tall from the base of the flower to the top of the flower.
5. Safety:
 - a. Must incite 0 total safety hazards for the crew whilst in operation, either by the method of power, operation, or form.
6. Structure:
 - a. Must stay on the pedestal, and not lean more than 0 degrees for the entirety of the play.
7. Reliability:
 - a. Must sustain 0 mechanical or operational issues or damages for at least 10 uses.
8. Power:
 - a. Must have access to at least 60v of power.
9. Cloche
 - a. Must not be enclosed by a cloche with an internal volume less than the volume of the rose.
10. Weight:
 - a. Must be lighter than 10lbs.

Constraints:

1. Cost:
 - a. Must not cost more than \$150.
2. Time:
 - a. First product must be done by February 20, 2024 for use by the stage crew.

By understanding these goals and limitations we feel we can create an effective device that will look like a rose as well as being functional to provide a magical appearance to the musical. Mrs Moody emphasized the rose must be as realistic as possible like having a narrow stem to give the audience the best experience.